

Aymen Rahmani

AI Engineering Student – LLMs, RAG & Agents

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Seeking a 6-month end-of-studies internship (PFE / Master's thesis) from February 2027 – LLMs, RAG & agents

Summary

Final-year engineering student specializing in **LLMs, RAG and agents**. Deployed an **agentic RAG assistant on Azure** with rigorous evaluation (0.93 correctness, Recall@5 0.94, 0/240 prompt injections succeeded, CI eval gate), a multi-modal triage agent, and a release-gate agent exposed over **MCP**. Two ML/CV internships shipped to production (FastAPI, ONNX).

Education

ESPRIT School of Engineering Sep 2024 – Jul 2027 (expected)
Engineering Degree (Diplôme d'Ingénieur, Master's level), Software & Data Science Engineering Ariana, Tunisia
Relevant coursework: Generative AI & NLP, Deep Learning, Machine Learning, MLOps, Probability & Statistics
Higher Institute of Computer Science (ISI) – Bachelor's Degree (Licence) in Embedded Systems Sep 2021 – Jul 2024

Experience

Computer Vision Intern Jun 2026 – Aug 2026
Supra Tech & Services Tunisia

- PCB defect inspection (**RT-DETR-L + PatchCore**): **77.8% mAP50:95 (+24 pts)** with EIoU + NWD losses.
- Real-time inference at 30 FPS (**ONNX, FastAPI**) with Eigen-CAM explanations for operators.

Machine Learning Intern Jul 2025 – Aug 2025
Web Media International Tunisia

- Email-risk classifier (**XGBoost**, cross-validation): **91% F1**, 90% accuracy on large-scale text data.
- Deployed as a **FastAPI** microservice integrated with a Spring Boot / MySQL backend.

Projects

Regulatory RAG Assistant deployed on Azure (GDPR, AI Act, NIS2, DORA) – personal project 2026
Python, FastAPI, Azure AI Search, Azure OpenAI, Container Apps, Terraform, GitHub Actions (OIDC), OpenTelemetry, pytest

- Three pipelines behind one streaming API: **hybrid RAG** (BM25 + dense, RRF), **GraphRAG** and **agentic RAG**; the agent reaches **0.93 correctness** vs 0.81 for the baseline (60 questions).
- Versioned **evaluation harness** (40+ runs): **Recall@5 0.73 → 0.94** from retrieval tuning; a **CI eval gate** blocks any PR that lowers answer quality; deploys with approval and rollback.
- Security and operations: auth, document-level security trimming, PII masking, **0 of 240 prompt-injection attacks succeeded**; OpenTelemetry tracing, about \$3.5 per 1,000 requests; 139 tests.

CrisisLink – Multimodal Flood Triage with LLM Agents (team of 6, final stage) Sep 2026 – now
n8n (53 nodes), Telegram, Groq (gpt-oss-120b, Qwen VL, Whisper), Gemini, Qdrant, multilingual-e5, Python

- Telegram bot (photo, voice note in Tunisian Derja / French / English, text, GPS): tool-using **Vision** and **Distress** agents, a **master triage agent** (P1–P4, which I own) with RAG on Qdrant, a **deterministic safety guardrail** and human escalation.
- Evaluations: distress extractor **89% → 100%** field accuracy (prompt V2 → V4, real transcripts); multilingual RAG **Hit@3 0.80 → 0.90**; vision: **0 critical photos under-graded** out of 29 labelled photos.

Autonomous Multi-Agent DevOps/MLOps Platform (team of 6, in progress) Sep 2026 – now
LangGraph, Kafka, gRPC/Protobuf, PostgreSQL, Qdrant, SonarQube, Sengrep, Trivy, Gitleaks, GitHub Actions, MCP, Docker

- Own the **Security & Quality agent**: tests, coverage, lint, CVEs (Trivy), secrets (Gitleaks) and SonarQube merged into one **ALLOW / WARN / BLOCK release verdict**.
- Public prototype wired into **CI (GitHub Actions)** and exposed as an **MCP server**; Kafka/gRPC platform where 6 agents fix incidents with human approval.

Certifications & Activities

NVIDIA DLI (5 certificates, 2025–26): deep learning, transformer-based NLP, LLM app development, prompt engineering, LLM evaluation

AWS Certified ML Engineer – Associate (in progress, exam Dec 2026) | **MASSAI AI Summer School** (ESPRIT, Jun 2025, 30 h)

Hackathons (Apr 2026): participated in “AI for Industrial Zones” (AFI & ESPRIT) and H12 Innovation 3.0 “AI Healing Gabès” (ESPRIT)

Technical Skills

LLMs & agents: LangChain, LangGraph, hybrid RAG (BM25 + dense, reranking), GraphRAG, agentic RAG, multi-agent systems, MCP, prompt engineering, Ollama, n8n
LLM eval & security: Recall@k, MRR, faithfulness, LLM-as-a-judge, CI eval gate, prompt-injection testing, PII masking
ML & data: Python, PyTorch, Transformers, Scikit-learn, XGBoost, Pandas, SQL, PostgreSQL, Qdrant, ChromaDB
Deployment & cloud: FastAPI, Docker, CI/CD (GitHub Actions, OIDC), Azure (AI Search, OpenAI, Container Apps), Terraform, OpenTelemetry, MLflow, Kafka
Other languages: C++, TypeScript
Languages: Arabic (native), **English (C1, certified)**, French (B2), German (B1–B2)